Fiscal Year:	FY 2025	**Task Last Updated:**	FY 10/31/2024
PI Name:	Yeung, Catherine Ph.D.		
Project Title:	Extended Culture of Kidney MPS and Organoids to Model Acute and Chronic Exposure to Drugs and Environmental Toxins		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:	NIH, BARDA, FDA	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	cathyy@uw.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	206-799-2374
Organization Name:	University of Washington		
PI Address 1:	4333 BROOKLYN AVE NE		
PI Address 2:			
PI Web Page:			
City:	Seattle	**State:**	WA
Zip Code:	98195	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA015N. Extended Longevity of 3D Tissues and Microphysiological Systems for Modeling of Acute and Chronic Exposures to Stressors
Start Date:	11/15/2022	**End Date:**	11/14/2026
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA HQ
Contact Monitor:	Koniges, Ursula	**Contact Phone:**	202-256-8786
Contact Email:	ursula.m.koniges@nasa.gov		
Flight Program:			
Flight Assignment:	Per the Research Announcement for this solicitation: this effort is a collaboration between NASA; the National Institutes of Health (NIH); the Biomedical Advanced Research and Development Authority (BARDA); and the Food and Drug Administration (FDA).		
Key Personnel Changes/Previous PI:	No changes in key personnel		
COI Name (Institution):	Bammler, Theodor Ph.D. (University of Washington) Freedman, Benjamin Ph.D. (University of Washington) Himmelfarb, Jonathan Ph.D. (University of Washington) Kelly, Edward Ph.D. (University of Washington) Thummel, Kenneth Ph.D. (University of Washington) Zelnick, Leila Ph.D. (University of Washington)		
Grant/Contract No.:	80ARC023CA001		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Kidney disease is a public health problem affecting 850 million people worldwide, 37 million people in the US adult population, and is the 9th leading cause of death. Loss of kidney function is often associated with chronic hypertension, diabetes, or exposure to pathogens, drugs, or environmental chemicals, and in many cases, takes many months or years to manifest clinically. Finding efficacious therapeutics and preventing disease progression require fundamentally new strategies, assays, and models of the kidney to improve patient outcomes. Until recently, the lack of human in vitro models that recapitulate critical aspects of kidney physiology, mimic the unique complexities of specific nephron segments, model disease heterogeneity, or assess reparative mechanisms in response to injury has hindered progress. Our team has approached the scientific challenges of studying kidney disease pathophysiology, novel therapeutic agents, and drug/ environmental toxicity by establishing two innovative human models: the proximal tubule kidney chip microphysiological system (PTEC MPS) and the iPSC (induced pluripotent stem cell)-derived kidney organoid. We have developed validation assays for short-duration (2-10 day) exposures in both ground-based laboratories and aboard the International Space Station. The goal of this project is to demonstrate models with extended (6 month) longevity to better understand the role of chronic stressor exposure on kidney disease initiation and progression. Based on previous studies, we hypothesize that human kidney systems will remain viable and functionally stable for at least 6 months and will respond physiologically to extended exposure to stressors. Phase I – Description and Tasks. Establish extended longevity and functionality (6 months) of kidney microphysiological systems and organoids. The contractor shall: • Determine longevity of the proximal tubule kidney chip (PTEC) MPS at 2.5, 5, and 6 months by employing cell viability (live/dead staining), metabolic function (conversion of 25(OH)VitaminD3 to 1a25(OH)2 VitaminD3), toxic response (resulting in KIM-1 [kidney injury molecule-1] expression), stable expression of immunomodulators (e.g., IL[interleukin]-6, IL-23A, TNF [tumor necrosis factor], IL-32), and global RNA transcript profile. • Determine longevity and stable function of kidney organoids at 2.5, 5, and 6 months by conducting observation of distinct tubules with specific multiple nephron and stromal lineages in segmented structures (immunocytochemical staining), and stable global single cell RNA transcript profile. Phase II – Description and Tasks. Test structural and functional outcomes after sustained exposure to drugs, environmental toxins, and pathogens and post-exposure cellular recovery. The contractor shall: • Determine functional outcomes of PTEC MPS after exposure to agents known to cause nephrotoxicity after 5 months exposure (PMB [polymyxin B] or other nephrotoxin) followed by 1 month recovery by evaluating structural derangement (de-polarization of transporter proteins), production of injury biomarkers (KIM-1, HO-1 [heme oxygenase-1], cystatin), and altered regulation of immunomodulators (e.g., IL-6, IL-23A, TNF, IL-32). • Evaluate kidney organoid responses to pathogenic stimuli and known nephrotoxins using fluorescent reporters for real-time detection of infection and injury in response to acute (3 days) and extended (5 months) exposure followed by 1 month of recovery.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Developing a kidney microphysiological system and kidney organoids with extended longevity will allow us to model the effects of stressors like drugs, environmental toxins, pathogens, and microgravity on kidney cell structure and function, and to determine if, and how, kidney cells can recover from stress. A better understanding of kidney injury and recovery from chronic stressor exposure can accelerate the development of better and safer medications and treatments to prevent and cure kidney diseases.
Task Progress:	Chips updates: Kidney chips were successfully cultured with >90% viability for more than 365 days. Functional tests (KIM-1 response to polymyxin B exposure and Vitamin D metabolism) and transcriptomics conducted at 6 months showed mixed results. IL-6 secretion and epithelial cell identity were stably maintained for 6 months; however, response to polymyxin B and Vitamin D metabolism were suppressed (but still present). This may be due to de-induction of response pathways after extended culture without physiological stimulus, or due to aging of the cells in culture. These studies were repeated and sample analysis is currently underway. Organoid updates: Previously, we were able to maintain viable human kidney organoids for six months. We have done additional analysis of kidney organoid structure, function, and gene expression (single cell RNA sequencing) from this experiment. We have also repeated the experiment successfully. The organoid team has now transitioned from Phase 1 to Phase 2 of the project. We are now focused on understanding how nephrotoxicity and pathogens affect kidney organoids in both acute and chronic settings. For pathogens (milestone 6), we have done a trial experiment for 60 days to test 3 different concentrations of IFN-gamma. For nephrotoxicity (milestone 7), we have optimized the doses of gentamicin and cisplatin in the acute setting and validated an antibody for KIM-1 detection that shows cellular damage localizing to the proximal tubular cells.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
Abstracts for Journals and Proceedings	Yeung CK, Abuhamad Y, Chalker B, Bain J, Himmelfarb J, Kelly EJ. "Use of kidney-on-chip to study long-term effects of microgravity on kidney structure and function." ISSRDC 2024 (the International Space Station Research & Development Conference), Boston, Massachusetts, July 29-August 01, 2024. Abstracts. ISSRDC 2024 (the International Space Station Research & Development Conference), Boston, Massachusetts, July 29-August 01, 2024. https://s3.amazonaws.com/amz.xcdsystem.com/4F14E44B-BC41-E69B-DFAF5A1B1627A0EA_abstract_File24568/FullBriefingPDF_94_0712022157.pdf , Jul-2024
Articles in Peer-reviewed Journals	Juliar BA, Stanaway IB, Sano F, Fu H, Smith KD, Akilesh S, Scales SJ, El Saghir J, Bhatraju PK, Liu E, Yang J, Lin J, Eddy S, Kretzler M, Zheng Y, Himmelfarb J, Harder JL, Freedman BS. "Interferon-γ induces combined pyroptotic angiopathy and APOL1 expression in human kidney disease." Cell Rep. 2024 Jun 25;43(6):114310. http://dx.doi.org/10.1016/j.celrep.2024.114310 ; PMID: 38838223; PMCID: PMC11216883 , Jun-2024